

Literacy in Coding

The medium chosen for this experience is coding, which I commenced learning in senior year of high school. This included learning how to understand basic syntax, observing the small intricacies of other writers when they code, and learning how to write it independently. This may sound familiar as it is strikingly similar to gaining literacy in a new language. The key difference being one's audience may also include a computer's CPU. The CPU in this case handles the logic a person gives it and sends a response back whether in the form of an error message or a successful compilation, similar to an audience trying to decrypt a writer's message.

The first step towards coding literacy was when my AP Computer Science teacher began explaining the way variables are written, and how methods and classes worked. At first, it felt completely foreign. Java was already an odd starting point for learning a programming language and understanding it was not any easier. This remained until our class began studying objects and learning about their implementation. Seeing how they all connect together as a person might create a class for an object which can hold the variables and methods mentioned above. From that point onwards the knowledge I gained felt incredibly intuitive rather than something recited merely from a lecture. That shift gave me the opportunity to read the logic of a person's code and become more engaged with the class. Soon enough Stack Overflow and outside media like YouTube became a viable resource for understanding statements and operations in the context of building objects.

After weaving past my first major stump, the next step was reading classmates' and other programmers' code. They all had subtle differences that functioned very similar to a writing style in this scenario, a term in this context that is also synonymous with its normal usage. It could show up in their brace and parenthesis placement, their naming conventions, comments, and also their logic used when performing operations. These were all silent indicators of how each person organized code in a way that made sense to them, and how they wanted to present it to others; be it messy or structured orderly. Thoughts like *'this seems cool, I want to implement this in my own writing at some point'* became more prominent. By this point my literacy in reading code had grown exponentially and made me think more critically about how I understand new subjects.

Eventually I reached the thought *'I bet I could try to write and implement this idea right now'*. That led to installing Java and an Integrated Development Environment (IDE) on a personal laptop at home and writing my own Java files. At first, writing objects and classes still posed a major challenge. Then learning how to use public and private variables effectively and dealing with some of Java's less appealing syntax when trying to implement methods on objects. After those early hiccups it became more imperative that the legibility of any syntax used was consistent. During the first time my Computer Science teacher read some of my code, the experience left some important notions that naming conventions made it difficult to follow at times. It needed to be ordered and precise, otherwise an attempt at communication became less convenient and almost arduous to read. It is not too dissimilar from writing a piece of media and struggling to convey a message when presented to an audience.

Literacy, in its actual form, seems to be bigger than just reading and writing; it can also include the means to think within a language, to understand not just what is being said but why it has been said the way it has. My journey through coding literacy followed that same kind of pattern. It began with decoding syntax like unfamiliar words on a page, then moved into reading others' logic like interpreting an author's voice and came together in writing code that others could follow. The ability to communicate through code, whether structured, legible, and intentional mirrors what it means to be literate in any medium of one's choice. Fluency was not gained at any single moment; it was built gradually with stepping stones along the way, through reading, imitating, and independent writing.